

Serena An

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Education

2026 – Present	Columbia University PhD Candidate in Mathematics.	New York, NY
2022 – 2026	Massachusetts Institute of Technology Bachelor of Science in Mathematics. Concentration in Theater Arts. <i>GPA: 4.9/5.0.</i>	Cambridge, MA
2021 – 2022	National Security Language Initiative for Youth (NSLI-Y) Took a gap year to study abroad with the selective NSLI-Y Korean Academic Year Program .	Seoul, South Korea

Selected Awards

2026	NSF Graduate Research Fellowship Program <i>National Science Foundation</i>	
2025	MIT Humanities Scholar <i>Burchard Scholars Program</i>	Cambridge, MA
2023	Honorable Mention (Top 100) <i>Putnam Mathematical Competition</i>	
2021	Gold Medal <i>European Girls' Mathematical Olympiad</i>	Kutaisi, Georgia (Online)
2018	TEDx Speaker <i>TEDxPioneerParkKids</i>	Brookings, SD

Research Experience

2026	<i>MIT MISTI Japan Research Exchange</i> Studied chromatic quasisymmetric functions under Tatsuyuki Hikita.	Kyoto, Japan
2025	<i>Imperial Summer Research Exchange</i> Studied group cohomology under Benjamin Briggs.	London, England
2024	<i>UMN Combinatorics and Algebra REU</i> Studied Postnikov-Stanley polynomials under Shiyun Wang and quiver representations under Kayla Wright.	Minneapolis, MN
2023	<i>MIT Summer Program for Undergraduate Research</i> Studied flip graphs under Elisabeth Bullock.	Cambridge, MA
2020	<i>MIT PRIMES-USA</i> Studied Grothendieck polynomials under Adela Zhang.	Online

Work Experience

2024 – Present	USA Deputy Leader <i>European Girls' Mathematical Olympiad</i> Coordinated olympiad papers and accompanied the USA EGMO team.	Europe (Varies)
2023 – 2026	Instructor <i>MIT Global Teaching Labs</i> Helped direct national math olympiad training camps.	South Africa; Ghana; Bhutan; Mexico
2022 – 2025	Teaching Assistant <i>Math Olympiad Summer Program</i> Graded olympiads and bridged the gap between instructor and student as an approachable role model.	Pittsburgh, PA

Research Reports

- 2025 Rank varieties for modules over quantum complete intersections
[pdf] under Benjamin Briggs
- 2024 Postnikov–Stanley polynomials are Lorentzian
[arXiv] with Katherine Tung and Yuchong Zhang
Algebraic Combinatorics, to appear.
- 2024 Newton polytopes of dual Schubert polynomials
[arXiv] with Katherine Tung and Yuchong Zhang
- 2024 Equality and saturated Newton polytopes of Postnikov–Stanley polynomials
[pdf] with Katherine Tung and Yuchong Zhang under Shiyun Wang and Meagan Kenney
- 2024 Mixed dimer model for exceptional type quiver representations and cluster algebras
[pdf] with Casey Appleton and Sogol Cyrusian under Kayla Wright and Elise Catania
- 2023 Flip graphs on self-complementary ideals of chain products
[arXiv] with Holden Mui under Elisabeth Bullock
- 2021 The Stembridge equality for skew stable Grothendieck polynomials and skew dual stable Grothendieck polynomials
[journal] with Fiona Abney-McPeck and Jakin Ng under Adela Zhang
Algebraic Combinatorics, Volume 5 (2022) no. 2, pp. 187-208.

Expository Writing

- 2025 Homotopy groups of spheres
[pdf] for MATH 231BR *Algebraic Topology II*.
- 2025 Faithful specializations of the Burau representation
[pdf] for 18.904 *Seminar in Topology*.
- 2024 Dyson Brownian motion as noncolliding Brownian motion
[pdf] for 18.676 *Stochastic Calculus*.
- 2024 Projective modules
[pdf] for 18.704 *Seminar in Algebra*.

Service and Mentorship

- Spring 2026 Undergraduate Assistant 18.900 *Geometry and Topology in the Plane*
- Fall 2025 Undergraduate Assistant 18.100A *Real Analysis*
- 2023 – 2025 Exec Member *Undergraduate Society of Women in Math*
- 2023 – 2026 Mentor *Undergraduate Society of Women in Math*
- 2023 – 2026 Mentor *Undergraduate Math Association*
- 2023 Tutor for 18.100A and 18.100Q *Undergraduate Math Association*
- 2022 – 2024 Problems Staff *Harvard-MIT Math Tournament*

Languages

English, Mandarin, Korean, Japanese